Troy Mayer

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Open to Summer 2027 internships in robot software and controls

EDUCATION

Purdue University, West Lafayette, IN Expected May 2028
B.S. Mechanical Engineering and B.S. Applied Statistics, Minor in Economics GPA: 3.32/4.00

SKILLS

- **Languages:** C++ (C++17), Python, MATLAB, SQL
- **Controls and estimation:** classical feedback control, IMC tuning, QP force control, error-state Kalman filtering, sensor fusion, observability analysis, rigid-body kinematics
- **Software and tools:** Linux, Git, CMake, GoogleTest, GitHub Actions CI, Eigen, MuJoCo, FreeRTOS, Flask, SQLite, pandas, NumPy, SciPy, PyTorch
- **Hardware and CAD:** KiCad, LabVIEW, DAQ, CATIA, Creo, FEA, CNC machining, GD&T

ROBOTICS EXPERIENCE AND PROJECTS

Boston Dynamics Jul 2026 – Present
Sensors and Perception Co-Op (Atlas) Waltham, MA

- Mapping Atlas impacts onto the robot mesh with **floating-base forward kinematics** from joint encoders and exact rigid-body velocity composition; estimating impact energy from operational-space effective mass and an OLS velocity fit.
- Designed a delta-v impact-severity metric that stays accurate when sensors miss an impact's peak; characterized its anti-alias filter on synthetic pulses and its noise floor with a null-control test.
- Building an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering about 28k impact events from about 900 robot test logs, used by engineers during testing; every change landed after bit-identical regression checks.

Legged Robot State Estimation and Control, *personal project, code* Aug 2026 – Oct 2026

- Built a **contact-aided error-state EKF in C++** (Eigen) with a Python reference, fusing a 1 kHz IMU and leg kinematics on a simulated Unitree Go1: 0.013 m/s velocity RMSE over a 60 s walk vs. 1.38 m/s IMU-only; ~20 µs median step.
- Verified with 26 GoogleTest cases (Jacobians against finite differences, C++/Python parity to 1e-13) and 0 warnings under -Wall -Wextra -Wpedantic; derived and numerically confirmed the four unobservable directions.
- Traced a ~4° lean to weak tilt observability on straight walks and fixed it with a 4 s startup turn; showed ~2 cm/s of foot slip doubles velocity error and removed a 6.6° yaw failure from 10 ms contact latency by gating on the gait schedule.
- Closed the loop in Python simulation: with the trot controller fed the estimate, 0 falls in five 60 s walks, within 0.004 m/s of ground-truth-fed tracking; added an MIT Cheetah 3-style QP force-balance controller (NNLS, 500 Hz).

Purdue ASME Biomechanics, Active Orthosis Apr 2024 – May 2026
Electronics Team Lead West Lafayette, IN

- Designed a **1 kHz PI motor velocity controller in C++17** for an ESP32-S3, IMC-tuned with feed-forward and anti-windup; co-simulated with a motor-to-forearm plant model: 0.8% speed error lifting at 15°/s vs. 98% feed-forward-only.
- Wrote the firmware around a pinned 1 kHz FreeRTOS task, an IRAM-safe encoder ISR, 10 latching fault monitors, a hardware watchdog, and a kill switch; 26 GoogleTest tests and an ESP32-S3 build run in GitHub Actions CI.
- Cut EMG-to-motion latency from 103 to 43 ms in closed-loop simulation on real recorded EMG, with a Bayesian amplitude estimator, a 20 ms onset confirm, and faster loop tuning.
- Led the 5-engineer electronics team; redesigned the 4-channel EMG board schematic in KiCad (INA333, right-leg drive, ISO7763 isolation), lifting emulated 11-class cross-day decoding from 31% to 87% on 115 days of real EMG.

Asteroid Hopping Robot, *independent research with Purdue University and UC3M (Madrid)* Oct 2026 – Present

- Building a three-legged hopping robot and a pendulum landing rig; designing push-off control to hit a target launch speed, knee impedance control for landing, and reaction-wheel attitude control, with hops sized in MuJoCo simulation.

ADDITIONAL EXPERIENCE

Caterpillar (via Purdue Data Mine), *Machine Learning Researcher* Jan 2026 – May 2026

- Trained LSTM and XGBoost models forecasting next-day paint consumption with Monte Carlo confidence intervals, achieving a 4.7% reduction in material usage; built Python pipelines that contributed to a 3.3% cut in cycle time.

Chortos Lab, Purdue, *Undergraduate Researcher* Aug 2025 – May 2026

- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm and LabVIEW/MATLAB data acquisition, and used them to formulate ionic dielectric gels with over 50% more force output.

Emerson, Discrete Automation (ASCO), *Mechanical and Manufacturing Engineer Intern* Jun 2025 – Aug 2025

- Designed and CNC-machined an adjustable spring canister for an industrial rotary valve in Creo, with stress calculations and GD&T drawings; assembled and tested 159 actuators and ran UL 429 HiPot tests on solenoid valves.